

Publication List

Wenhan Cao

Updated October 2026

Published and Accepted Papers (* denotes equal contribution)

- [1] W. Cao, X. Chen, S. Wang, and L. Zhao. “How Much Information is Needed for Accurate Kalman Filtering?” In: *Advances in Neural Information Processing Systems (NeurIPS)*. Accepted. 2026.
- [2] W. Cao, T. Zhang, Z. Sun, C. Liu, S. S.-T. Yau, and S. E. Li. “Nonlinear Bayesian Filtering With Natural Gradient Gaussian Approximation”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 48.7 (2026), pp. 8581–8597.
- [3] W. Cao, S. Liu, C. Liu, Z. He, S. S.-T. Yau, and S. E. Li. “Convolutional Bayesian Filtering”. In: *IEEE Transactions on Automatic Control* (2026), pp. 1–8.
- [4] K. Zhou, Z. Wen, Z. Zhuo, Z. Yan, P. Wu, S. Li, H. Gao, K. Ding, W. Cao, W. Pan, and C. Liu. “CoINS: Counterfactual Interactive Navigation via Skill-Aware VLM”. In: *IEEE Transactions on Automation Science and Engineering* (2026).
- [5] T. Zhang, W. Cao, and S. E. Li. “Natural Gradient Gaussian Approximation Filter with Positive Definiteness Guarantee”. In: *2026 American Control Conference*. 2026, pp. 3224–3229.
- [6] S. Liu*, W. Cao*, C. Liu, Z. He, T. Zhang, Y. Wang, and S. E. Li. “One Filters All: A Generalist Filter For State Estimation”. In: *Advances in Neural Information Processing Systems*. Vol. 38, Main Conference. Curran Associates, Inc., 2025, pp. 21276–21302.
- [7] T. Zhang, W. Cao, C. Liu, T. Zhang, J. Li, and S. E. Li. “Robust State Estimation for Legged Robots With Dual Beta Kalman Filter”. In: *IEEE Robotics and Automation Letters* 10.8 (2025), pp. 7955–7962.
- [8] W. Cao, C. Liu, Z. Lan, S. E. Li, W. Pan, and A. Alessandri. “Robust Bayesian Inference for Moving Horizon Estimation”. In: *Automatica* 173 (2025), p. 112108. ISSN: 0005-1098.
- [9] T. Zhang, W. Cao, C. Liu, F. Zhang, W. Wu, and S. E. Li. “NANO-SLAM: Natural Gradient Gaussian Approximation for Vehicle SLAM”. In: *2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)*. 2025, pp. 4014–4019.
- [10] W. Cao, A. Capone, R. D. Yadav, S. Hirche, and W. Pan. “Computation-Aware Learning for Stable Control with Gaussian Process”. In: *Proceedings of Robotics: Science and Systems*. Delft, Netherlands, July 2024.
- [11] W. Cao and W. Pan. “Impact of Computation in Integral Reinforcement Learning for Continuous-Time Control”. In: *The Twelfth International Conference on Learning Representations*. 2024.
- [12] S. Liu, W. Cao, C. Liu, T. Zhang, and S. E. Li. “Convolutional Unscented Kalman Filter for Multi-Object Tracking with Outliers”. In: *IEEE Transactions on Intelligent Vehicles* 10.5 (2025), pp. 3113–3123. DOI: [10.1109/TIV.2024.3446851](https://doi.org/10.1109/TIV.2024.3446851).
- [13] J. Duan, W. Cao, Y. Zheng, and L. Zhao. “On the Optimization Landscape of Dynamic Output Feedback Linear Quadratic Control”. In: *IEEE Transactions on Automatic Control* 69.2 (2024), pp. 920–935.
- [14] W. Cao, C. Liu, Z. Lan, Y. Piao, and S. E. Li. “Generalized Moving Horizon Estimation for Nonlinear Systems with Robustness to Measurement Outliers”. In: *2023 American Control Conference*. 2023, pp. 1614–1621.
- [15] W. Wu, C. Liu, W. Cao, C. Chen, H. Li, and S. E. Li. “Multi-Sensor Calibration System for Mobile Robots”. In: *2023 7th CAA International Conference on Vehicular Control and Intelligence*. 2023, pp. 1–7.
- [16] W. Wang, Y. Zhang, J. Gao, Y. Jiang, Y. Yang, Z. Zheng, W. Zou, J. Li, C. Zhang, W. Cao, G. Xie, J. Duan, and S. E. Li. “GOPS: A General Optimal Control Problem Solver for Autonomous Driving and Industrial Control Applications”. In: *Communications in Transportation Research* 3 (2023), p. 100096.

- [17] W. Cao, J. Duan, S. E. Li, C. Chen, C. Liu, and Y. Wang. “Primal-Dual Estimator Learning Method with Feasibility and Near-Optimality Guarantees”. In: *2022 IEEE 61st Conference on Decision and Control*. 2022, pp. 4104–4111.
- [18] J. Duan, W. Cao, Y. Zheng, and L. Zhao. “On the Optimization Landscape of Dynamic Output Feedback: A Case Study for Linear Quadratic Regulator”. In: *2022 IEEE 61st Conference on Decision and Control*. 2022, pp. 4040–4045.
- [19] Y. Yin, S. E. Li, K. Tang, W. Cao, W. Wu, and H. Li. “Approximate Optimal Filter Design for Vehicle System through Actor-Critic Reinforcement Learning”. In: *Automotive Innovation 5.4* (2022), pp. 415–426.
- [20] Z. Gu, Y. Yang, J. Duan, S. E. Li, J. Chen, W. Cao, and S. Zheng. “Belief State Separated Reinforcement Learning for Autonomous Vehicle Decision Making under Uncertainty”. In: *2021 IEEE International Intelligent Transportation Systems Conference*. 2021, pp. 586–592.
- [21] W. Cao, J. Chen, J. Duan, S. E. Li, Y. Lyu, Z. Gu, and Y. Zhang. “Reinforced Optimal Estimator”. In: *IFAC-PapersOnLine 54.20* (2021). Modeling, Estimation and Control Conference 2021, pp. 366–373.
- [22] J. Li, S. E. Li, K. Tang, Y. Lv, and W. Cao. “Reinforcement Solver for H-infinity Filter with Bounded Noise”. In: *2020 15th IEEE International Conference on Signal Processing*. Vol. 1. 2020, pp. 62–67.
- [23] L. Xin, S. E. Li, P. Wang, W. Cao, B. Nie, C.-Y. Chan, and B. Cheng. “Accelerated Inverse Reinforcement Learning with Randomly Pre-sampled Policies for Autonomous Driving Reward Design”. In: *2019 IEEE Intelligent Transportation Systems Conference*. 2019, pp. 2757–2764.

Preprints and Submissions

- [24] W. Cao, K. Yan, Y. Feng, R. S. M. Goh, and J. Zhou. *HIRE: Hippocampal Replay for Test-Time Agentic Reinforcement Learning*. Under review at ICLR 2027. 2026.
- [25] K. Yan, W. Cao, S. Wang, Y. Yang, Wu Junke, X. Chen, and L. Zhao. *Generating Parallel Worlds of a Flight*. Under review at ICLR 2027. 2026.
- [26] K. Yan, X. Chen, W. Cao, and L. Zhao. *Evaluating OOD Actions in Offline RL via Quantile Advantage*. Under review at ICLR 2027. 2026.
- [27] W. Cao, K. Yan, K. Wang, X. Chen, and L. Zhao. *Trajectory Editing as Posterior Inference for Offline Reinforcement Learning*. Under review at ICLR 2027. 2026.
- [28] X. Chen, K. Yan, D. Chen, H. Li, W. Cao, and L. Zhao. *Learning Robotic Skills from Offline Preference-based RL*. Under review at ICLR 2027. 2026.
- [29] R. Huang, X. Chen, W. Cao, L. Li, Y. Xiao, Y. Gao, Y. Shi, and L. Zhao. *Agile Modular Aerial Robot Load Transport with Arbitrary Configurations*. Under review at IEEE/ASME Transactions on Mechatronics (T-MEC). 2026.
- [30] H. Gao, W. Cao, I. H. U, K. Zhou, Y. Sun, A. Alessandri, and C. Liu. *Chance-Constrained Gaussian Filtering*. Under review at Automatica. 2026.